

30-Day Interview Preparation Roadmap

Target Role: Generative AI Engineer | AI/ML Engineer
Focus: Resume-Based Project Preparation

Week 1 – Python & Backend Foundation

Python Core

- Variables
- Data Types
- Operators
- Control Flow
- Functions
- Modules & Packages

Object-Oriented Programming

- Class & Object
- Constructor
- Inheritance
- Polymorphism
- Encapsulation
- Abstraction
- Decorators

Exception Handling

- try / except / else / finally
- raise
- Custom Exceptions

File Handling

- Read
- Write
- CSV
- JSON
- Pickle

JSON

- Serialization
- Deserialization

HTTP Basics

- Client-Server
- Request & Response
- Methods
- Headers
- Status Codes

REST APIs

- GET
- POST
- PUT
- DELETE
- JSON Response

FastAPI

- Routing
- Request
- Response
- Pydantic
- Async/Await
- Dependency Injection

Flask

- Routing
- Request
- Response
- Sessions
- File Upload

NumPy

- Arrays
- Indexing
- Broadcasting
- Vectorization

Pandas

- DataFrame
- Series
- Filtering
- GroupBy
- Merge

Week 2 – AI & Deep Learning

Machine Learning

- Supervised
- Unsupervised
- Train/Test Split
- Overfitting
- Underfitting

Neural Networks

- Neuron
- Layers
- Activation Functions

Gradient Descent

- Learning Rate
- Backpropagation

Loss Functions

- Cross Entropy
- MSE

Optimizers

- SGD
- Adam

CNN

- Convolution
- Filters
- Padding
- Stride
- Pooling
- Dense
- Softmax

Transfer Learning

- Feature Extraction
- Fine Tuning
- Frozen Layers

TensorFlow

- Tensor

- Sequential API
- Functional API
- Model Loading
- Prediction

Keras

- Layers
- Compile
- Fit
- Evaluate
- Predict

OpenCV

- Image Loading
- Resize
- Normalization
- Color Conversion

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

Week 3 – LLM Fundamentals

Transformer

- Self Attention
- Multi-Head Attention
- Encoder
- Decoder

LLM Basics

- Tokens
- Tokenization
- Context Window
- Temperature
- Top-K
- Top-P

Prompt Engineering

- System Prompt
- User Prompt
- Few-Shot
- Zero-Shot
- Chain of Thought

Google Gemini API

- Authentication
- API Calls
- Vision
- Streaming
- Session Management

Groq API

- Model Selection
- Streaming
- Rate Limits

Streaming Responses

Multi-turn Conversations

Context Management

Week 4 – RAG

LangChain

- Architecture
- Chains
- Prompt Templates
- Output Parser

RAG

- Architecture
- Workflow
- Retrieval Pipeline

Document Loaders

Text Splitters

Chunking

- Chunk Size
- Chunk Overlap

Embeddings

- Sentence Embeddings
- Embedding Models

FAISS

- Index
- Similarity Search
- Top-K Retrieval

ChromaDB

- Collections
- Add Documents
- Query Documents

Semantic Search

Retriever

Context Injection

Prompt Construction

Response Generation

Week 5 – Deployment & Production

Git

- init
- add
- commit
- push
- pull
- branch
- merge

GitHub

Docker (Basic)

- Images
- Containers
- Dockerfile
- Basic Commands

Vercel

- Deployment
- Environment Variables

Environment Variables & Secrets

Logging

Debugging

Error Handling

JavaScript (Talk2Tube Only)

- ES6 Basics
- Fetch API
- Async/Await

Chrome Extension APIs

- Manifest.json
- Content Script
- Background Script
- Message Passing
- Chrome Storage

Final Phase – Project Mastery

PlantCareAI

- Workflow
- Image Preprocessing
- CNN
- Transfer Learning
- TensorFlow/Keras
- OpenCV

- Flask
- Gemini API
- REST API
- Deployment

SummarAlze

- RAG Pipeline
- PDF Processing
- Chunking
- Embeddings
- FAISS
- Retriever
- LangChain
- FastAPI
- Groq API
- Response Generation

Talk2Tube

- Transcript Extraction
- Chunking
- Embeddings
- ChromaDB
- LangChain
- FastAPI
- JavaScript
- Chrome Extension APIs
- Workflow

Final Revision

- Resume Review
- Project Explanation
- Cross Questions
- Debugging
- Mock Interview
- HR Questions